

EASYLAW: END-TO-END SYSTEM SPECIFICATIONS

Complete Technical & Functional Specification
Document for On-Demand Legal Consultation

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1. Executive Abstract & Scope

****EasyLaw**** is a production-ready, on-demand legal consultation platform designed to bridge the gap between people seeking legal advice and verified advocates. By using real-time call bridging, database session tracking, and SMS/WhatsApp re-engagement engines, EasyLaw provides a streamlined, secure experience for legal consulting.

The scope of this project covers three major application environments: the client web client, the lawyer management portal, and the central administrative dashboard. Through integrations with telephony services (Exotel) and payment gateways (Razorpay), the platform automates lead assignments, masks phone numbers to protect user privacy, and handles system failures without human intervention.

2. Complete User Dashboard Workflows

This section outlines the operational capabilities, tabs, and interactive structures of each portal.

2.1. Client Dashboard Operations

The client portal guides users from registration to post-call feedback in a clear, step-by-step flow:

- **Onboarding & Phone Validation:** The client logs in using OTP authentication. The user profile is stored securely, keeping track of active sessions and draft states.
- **Client Intake Form:** Users enter their details, choosing their location (Delhi, Gautam Buddha Nagar, or Ghaziabad) and legal category (e.g., Family & Marriage, Property & Registry, Criminal & Police, Corporate, Supreme Court, Cyber Fraud, Employment, Consumer, or Other). They also provide a case brief and choose between "ASAP (60 min)" or "Later Today" callback windows.
- **State-Based Consultation Tracker:** Once submitted, the client is redirected to a dynamic tracker page. This tracker polls the backend database every 5 seconds to show updates in real-time, showing steps like: Pending Payment → Searching for Advocate → Advocate Assigned (Matches name/bar status) → Bridge Call Connected → Consultation Resolved.
- **Ratings & Feedback Console:** When a consultation ends, the client's screen shows a feedback modal. Users can rate their advocate (1 to 5 stars) and write reviews, which immediately update the lawyer's metrics.

2.2. Lawyer Dashboard Operations

The lawyer console is designed to help advocates handle incoming calls and track their performance easily. It features 5 main sections:

- **Availability Toggle Console:** A simple switch that changes the lawyer's status between "Online" and "Offline" in the database. When marked online, the lawyer becomes visible to the auto-matching engine for incoming client leads.
- **Calls Hub (Active Leads Panel):** Shows incoming, high-priority consultation request cards. Each card displays the client's name, category, and issue description. The lawyer can click **"Accept Case"** to lock the lead and start the bridged phone call.
- **Cases History Archive:** A detailed history of all completed, cancelled, or forwarded consultations, listing client details, total call durations, notes, and user ratings.
- **Earnings Console & Payout Logs:** A financial tab displaying total earnings, pending payouts, commission splits, and bank settlement histories.
- **Profile & KYC Settings:** Shows state bar council documents, bar membership ID numbers, verification status, and practice category specializations.

2.3. Administrator Panel Operations

The Admin Panel serves as the central control tower for managing the system, handling exceptions, and overseeing operations:

- **Global Lead Management Directory:** Lists all client requests in the database. Admins can edit details, update lead priorities, or cancel active sessions.
- **Manual Match Override Console:** Allows admins to bypass the automated matching system. If a client requires a specific lawyer or if the matching engine times out, the admin can manually pair the lead and trigger the Exotel call bridge.
- **Advocate KYC Review Panel:** A queue displaying new lawyer applications. Admins review Bar Council enrollment certs and ID documents before approving or blocking access.
- **Telephony Logs & Call Auditing Registry:** Monitors Exotel Call Detail Records (CDRs), recording files, call durations (in seconds), and webhook response payloads for tracking.

3. The Lead Lifecycle & State Machine

The status of a consultation request is tracked using the status property on the Lead table. The table below shows the states a lead can move through based on client, lawyer, admin, and background SLA actions.

State	Trigger Condition	System Event / Action	Client View	Lawyer View
NEW	Client submits the intake form.	Database record created; draft state is cleared from localStorage.	Redirects to Payment Page.	Unavailable (not matched yet).
PAID	Payment gateway payment is success.	Razorpay webhook processes event; updates lead state; triggers auto-match search.	Shows "Finding matched lawyer..." loading screen.	Lead card appears in the "Calls Hub" for matched, online lawyers.
ASSIGNED	Lawyer clicks "Accept Case" in their dashboard.	Locks lead to this lawyer; stops matching notifications; calls Exotel bridge API.	Spinner updates to show Lawyer's name, Bar ID, and ratings.	Initiates call bridge loading indicator; dials lawyer's phone.
COMPLETED	Exotel call hangs up, logging a valid duration.	Exotel webhook reports call completion; updates lead status; unlocks lawyer.	Shows ratings and review modal.	Returns dashboard to active state; displays updated earnings.
ESCALATED	No lawyer accepts ASAP case within 15 mins.	SLA cron shifts lead to Flow 4; activates alarm banner on Admin Panel.	Shows "Searching taking longer than usual... hold on" status update.	Lead disappears from lawyer dashboard Calls Hub.
CANCELLED	Client declines, payment fails, or admin manually voids.	Releases resources; deletes sessions; records logs.	Shows "Booking Cancelled/Refunded" alert.	Lead disappears.

4. Telecom Call Routing Mechanics (Exotel Dialing Workflow)

To protect privacy and comply with data safety laws, EasyLaw keeps client and lawyer phone numbers masked using a secure bridged call system. Below is the detailed workflow of the call process:

Exotel Call Bridging Workflow:

1. When a lawyer accepts a case, the server makes an HTTP POST call to the **Exotel Outbound Call API**, sending the lawyer's number, the client's number, the caller ID (virtual number), and callback webhook URLs.
2. Exotel first calls the **Lawyer's phone number**.
3. If the lawyer declines or doesn't answer within 30 seconds, the call is terminated, and the system resets the lead state to look for a new match.
4. Once the lawyer answers, the Exotel system plays a brief voice confirmation: *"Connecting you to your client on EasyLaw..."*
5. The system then automatically dials the **Client's phone number**.
6. When the client answers, both lines are joined. The call is routed through Exotel's virtual bridge number, masking the private numbers of both users.
7. After the call ends, Exotel sends a POST request (webhook callback) to our server with the call record, including the duration, start time, end time, and a link to the recording.

5. System SLAs & Background Automation (Cron Engines)

The platform uses backend services running every 5 minutes to handle timeouts, follow up on incomplete requests, and manage system escalations automatically.

5.1. Incomplete Session Drop-off Engine

If a guest or client starts filling out the intake form but drops off before completing the registration or payment:

- **T+30 Minutes (Nudge 1):** The system detects the inactive state and sends an automated WhatsApp message via the Meta API, inviting the user to resume their draft.
- **T+4 Hours (Nudge 2):** If still incomplete, a final reminder is sent stating that advocates specializing in their category are active and waiting to consult.
- **T+24 Hours (Expiry):** If the user does not respond, the session is marked as expired and cleared from active queues. The draft remains cached in their local browser storage for easy manual retrieval.

5.2. ASAP Call Match & Escalation SLA

For high-priority cases (Flow 2 - ASAP):

- ****T+15 Minutes (No acceptance):**** If a lead is created but no online lawyer clicks "Accept" within 15 minutes, the system immediately escalates the lead to ****Flow 4 (Emergency/Escalated)**** in the Admin Panel.
- ****T+5 Minutes (No answer on call):**** If a lawyer accepts a case but fails to answer the incoming Exotel bridge call, the case is immediately returned to the queue, and an alert is sent to the admin.
- ****Admin Override:**** The admin is alerted on their dashboard via audio/visual banners and can manually assign the lead to any available lawyer in the database, bypassing the matching engine.

6. Regulatory Compliance & Data Safety

- ****Digital Personal Data Protection (DPDP) Act, 2023:**** EasyLaw complies with personal data standards. Phone numbers are encrypted in transit and masked during call routing. Access to call logs and details is restricted to authorized admin profiles.
- ****TRAI TCCCPR 2018:**** All transactional messages, OTP validations, and call alerts conform to TRAI guidelines. They are routed using pre-registered DLT headers and template IDs to prevent delivery failures.